Smart Menu System — Technical & Architectural Analysis

Project Location: /Users/prabhmehardhalio/Desktop/smart-menu-main | **Date:** September 9, 2026

Executive Summary

This document provides a comprehensive software engineering evaluation, architectural review, and technical roadmap for the **Smart Menu Project**. The platform features an AI-driven digital menu system powered by a FastAPI backend (supporting Neon PostgreSQL and SQLite) and a React + Vite frontend.

1. Project Overview & Current Implementation Status

- **AI Menu Extraction:** Parses menu images/PDFs via Groq LLM into structured JSON categories & items, enriched with Unsplash dish photography.
- **Cryptographic Table QR Management:** Issues table-specific JWT tokens signed with ACID_SECRET_KEY, supporting on-demand security resets.
- **Geofencing & Proximity Security:** Validates customer GPS coordinates (lat, lon) within a 200m radius of the restaurant location.
- **Customer Digital Ordering:** Full mobile menu UI with search, category filtering, cart management, customer detail capture, special notes, and checkout.
- **Live Tracking & Invoicing:** Customer-facing order tracking timeline with automatic PDF invoice download (jsPDF).
- **Kitchen & Admin Dashboard:** Complete operational dashboard featuring real-time KPI metrics, kitchen kanban queue, CRM, and settings.

2. What Needs to be Implemented (Technical Roadmap)

The following core features must be implemented to convert the project into a commercial-grade SaaS platform:

Feature Area	Current Implementation Gap	Required Technical Implementation
Payment Gateway	Orders transition to 'transmitted' directly without collecting payment.	Integrate Stripe / Razorpay SDKs with backend webhooks and payment status fields (paid, unpaid).
Admin Auth & RBAC	Admin routes and backend APIs lack user login authentication.	Implement JWT/Session admin login with bcrypt password hashing and Role-Based Access Control.
Real-Time WebSockets	Frontend relies on 10-second HTTP polling for order status updates.	Replace polling with WebSockets or Server-Sent Events (SSE) for instant kitchen alerts.
Item Availability ('86d')	Sold-out dishes cannot be disabled dynamically by kitchen staff.	Add is_available flags in database schema with live toggle switches on kitchen dashboard.
Item Modifiers & Add-ons	Dishes have fixed single prices without variants or options.	Expand schema for size variants (S/M/L), toppings, and dietary tags (Vegan, Jain, Halal).
Call Waiter Feature	No digital button for customers to ping staff for table service.	Add 'Call Waiter / Request Assistance' button sending push alerts to staff dashboards.

Multi-Language (i18n)	Menu text is locked to the original extracted OCR language.	Integrate i18n auto-translation for international tourist languages.

3. Payment Gateway Strategy & Selection

Recommended Gateways: Stripe (Global) & Razorpay (India Market)

Stripe (Global Markets): Crucially supports **Apple Pay** (Safari iOS) and **Google Pay** (Chrome Android) via Web Payment APIs. Diners scanning QR codes at tables do not want to manually type 16-digit credit card numbers on a web page. Stripe Elements enables instant 2-second authorization via Face ID or Touch ID.

Razorpay (India Market): Natively supports **UPI (Google Pay, PhonePe, Paytm, BHIM)**, which accounts for over 80% of digital dining payments in India. Allows seamless mobile app intent switching directly from the web browser.

4. Architectural Evaluation: Mobile Web App vs Mobile Application

Recommendation: Mobile Web Application (Progressive Web App - PWA) for Customers

- **Zero Customer Friction:** Diners scan QR code → menu opens instantly in mobile browser (< 1 second). Requiring an App Store download (50MB+) causes >90% customer drop-off.
- **Native Camera QR Scanning:** iOS/Android camera apps parse table URLs directly without installing any software.
- **Instant Mobile Web Payments:** Apple Pay and Google Pay operate natively inside mobile web browsers via Payment Request APIs.
- **Single Codebase Maintenance:** One React + Vite codebase serves customer phones, desktop admin, and kitchen display tablets.

5. Visual Theme Evaluation & Selection

Recommendation: Dynamic Multi-Theme Engine
The current codebase features an **Acid Maximalism / Dark Cyberpunk** theme (Acid Lime #D6FF00, Acid Cyan #00F0FF, Acid Pink #FF00FF, Jet Black #050505). While ideal for nightlife bars, sports lounges, and gaming cafes, fine dining and family restaurants require a warm, soothing minimalist aesthetic. Enabling dynamic theme selection in Admin Settings ensures universal market fit across all restaurant types.